Q1.

During pregnancy a woman's body increases in mass.
The table shows the average increase in mass in some parts of the body during pregnancy.

part	increase in mass during pregnancy (kg)
foetus	3.6
uterus	0.9
placenta	0.7
red blood cells	0.2
amniotic fluid	0.9
breast tissue	0.4
fat	3.9

(a) Explain why the mass of the placenta increases as the foetus develops.

.....

.....

.....

2 marks

(b) Pregnant women need to make sure they have plenty of iron in their diet.
Use information in the table to explain why they need extra iron.

.....

.....

1 mark

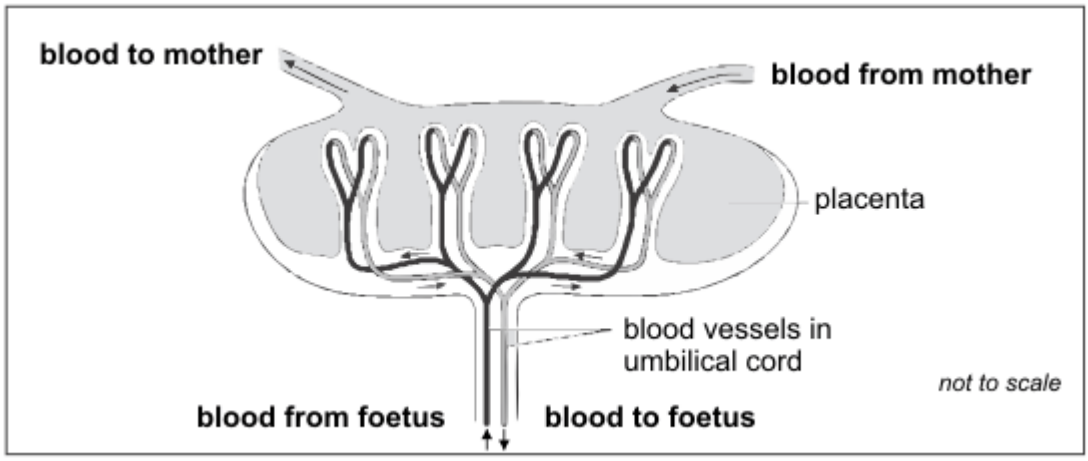
(c) The foetus is **not** part of a woman's body before she becomes pregnant.

Which **two** other parts from the table are **not** present in her body before she becomes pregnant?

..... and .....

1 mark

(d) (i) The diagram shows the blood supply in the placenta and umbilical cord.

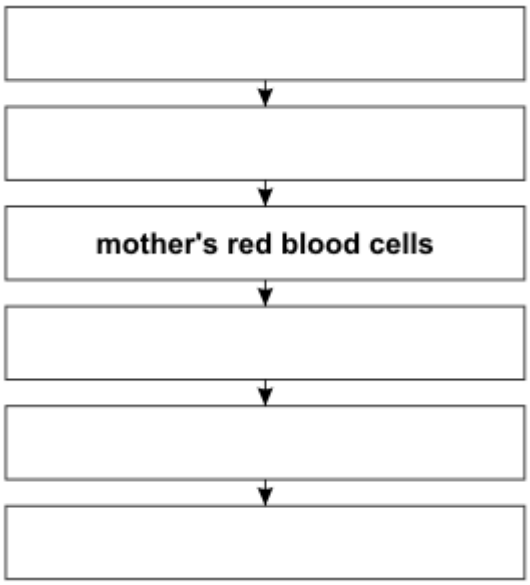


not to scale

When the mother breathes, oxygen and other gases pass to the foetus.

Complete the flow diagram below to show how oxygen passes from the mother to the foetus. Use **all** the words from the list below.

lungs	**umbilical cord**	**blood of foetus**	**windpipe**	**placenta**



2 marks

(ii) When a pregnant woman breathes in cigarette smoke, carbon monoxide gas combines with some of her red blood cells.

How could this harm the foetus?

.....

.....

1 mark
maximum 7 marks

Q1.

(a) any **one** from

- the placenta provides **or** passes oxygen **or** food **or** gets rid of waste
  accept responses that refer to a baby rather than a foetus

- the foetus needs oxygen **or** food **or** to remove waste
  'the foetus needs blood from the placenta' is insufficient

1 (L7)

- as the foetus develops it needs more (food **or** oxygen from the placenta)
  'it needs or provides more oxygen' is insufficient as the meaning is ambiguous
  *accept, for two marks, 'the placenta provides more oxygen **or** food **or** waste removal'*
  *accept, for two marks, 'the foetus needs **more** oxygen **or** food **or** waste removed as it grows'*
  *a two mark answer must include one of the first two marking points and a reference to the foetus needing **more** food or oxygen from the placenta*

1 (L7)

(b) any **one** from

- the number **or** mass of red blood cells increases
  accept 'they need (to make) more red blood cells'
  'the mass of blood increases' is insufficient

- iron is needed for the formation of red blood cells
  accept 'for haemoglobin'
  'the number of blood cells increases' is insufficient
  'for red blood cells' is insufficient

1 (L7)

(c) placenta
amniotic fluid
both answers are required for the mark
'umbilical cord' is insufficient
answers may be in either order

1 (L7)

(d) (i)
- windpipe
  lungs
  mother's red blood cells
  award a mark for 'windpipe' and 'lungs' in the first places in the correct order

- placenta
  umbilical cord
  *accept 'umbilical' **or** 'cord'*

blood of foetus

accept 'umbilical cord' and 'blood of foetus' in either order

award a mark for 'placenta', 'umbilical cord' and 'blood of foetus' in the last three places in an appropriate order

*all **five** answers are required for two marks*

2 (L7)

(ii) any **one** from

- less oxygen passes through the placenta
- less oxygen in the foetus' blood

  accept 'the foetus' heart beat rises'
  ***or** 'the foetus' blood pressure goes up'*
- smaller **or** brain damaged foetus

  accept 'less oxygen in the mother's blood'

  accept 'less oxygen for the foetus'

  *do **not** accept 'no oxygen'*
- baby may be premature

  accept responses that refer to a baby rather than a foetus

  accept 'the red blood cells of the foetus will contain carbon monoxide'

  *do **not** accept 'only carbon monoxide passes through the placenta'*
  'it passes to the foetus' blood' is insufficient
  'it poisons the foetus' is insufficient

1 (L7)

[7]